

The General Converses for Two-Sided Orthogonality Maps

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Abstract

Fadaee and Ghahramani ask whether the converses of their Results 3.2 and 3.3 hold for general C^* -algebras. We give direct affirmative proofs. If $\delta = d + \eta(\cdot)$ with d a derivation and $\eta \in \mathcal{Z}(A^{**})$, then the two zero-product identities are just the Leibniz rule plus central terms proportional to ab or ba . If $\delta = d_1 + \eta(\cdot) = d_2 + (\cdot)\eta$ with d_1, d_2 star derivations, use the left representation for the identity at $ab^* = 0$ and the right representation for the identity at $b^*a = 0$. Thus both structural conclusions are also sufficient. The second converse does not require the additional condition $\text{Re } \eta \in \mathcal{Z}(A^{**})$.

Status: candidate full result, likely valid. The final open question of arXiv:1907.03594 is answered affirmatively in the same bidual-valued generality as its two structural results. Novelty is plausible but not certified.

1 Exact source question

Let A be a C^* -algebra, canonically embedded in its bidual A^{**} . The source proves the following one-way implications [1].

- Result 3.2: if a continuous linear map $\delta : A \rightarrow A^{**}$ satisfies

$$ab = ba = 0 \implies a\delta(b) + \delta(a)b = b\delta(a) + \delta(b)a = 0,$$

then

$$\delta(a) = d(a) + \eta a, \quad d : A \rightarrow A^{**} \text{ a continuous derivation, } \eta \in \mathcal{Z}(A^{**}). \quad (\text{R3.2})$$

- Result 3.3: for unital A , if $\delta : A \rightarrow A^{**}$ satisfies

$$ab^* = b^*a = 0 \implies a\delta(b)^* + \delta(a)b^* = \delta(b)^*a + b^*\delta(a) = 0,$$

then there are continuous star derivations $d_1, d_2 : A \rightarrow A^{**}$ and $\eta \in A^{**}$, with $\text{Re } \eta \in \mathcal{Z}(A^{**})$, such that

$$\delta(a) = d_1(a) + \eta a = d_2(a) + a\eta. \quad (\text{R3.3})$$

After proving converses under extra hypotheses in Proposition 3.4, the source asks whether the converses of Results 3.2 and 3.3 hold in general:

□

Note that in Proposition 3.4 the converses of Results 3.2 and 3.3 hold, but we do not know the converses of these results are true or not, in general.

Here a star derivation is a derivation satisfying $d(x^*) = d(x)^*$. Products below are the canonical left and right actions of A on A^{**} .

2 Idea of the proof

Proof Intuition

The requested implications are formal once one chooses the correct decomposition. In (R3.2), the derivation part combines by Leibniz into $d(ab)$ or $d(ba)$, while centrality moves η across an element of A ; all remaining terms contain the assumed zero product.

In (R3.3), there are two decompositions for a reason. The expression $a\delta(b)^* + \delta(a)b^*$ is compatible with the left-multiplier form $d_1 + \eta(\cdot)$ and collapses to $d_1(ab^*)$. The other expression is compatible with the right-multiplier form $d_2 + (\cdot)\eta$ and collapses to $d_2(b^*a)$. No commutation of η is needed.

3 Full resolution

Theorem 1. *The converses of Results 3.2 and 3.3 of [1] are true. More precisely:*

1. *Let A be any C^* -algebra. Every map of the form (R3.2) satisfies both two-sided zero-product identities of Result 3.2.*
2. *Let A be any C^* -algebra for which data in (R3.3) are given. Then δ satisfies both star-orthogonality identities of Result 3.3. This implication needs neither unitality nor $\text{Re } \eta \in \mathcal{Z}(A^{**})$.*

Consequently the hypotheses and structural conclusions in each source result are equivalent (within the hypotheses under which the source proves the forward implication).

Proof. For the first assertion, suppose $\delta(x) = d(x) + \eta x$, where d is a derivation and $\eta \in \mathcal{Z}(A^{**})$. If $ab = ba = 0$, associativity and centrality give

$$\begin{aligned} a\delta(b) + \delta(a)b &= ad(b) + d(a)b + a\eta b + \eta ab \\ &= d(ab) + 2\eta ab = 0, \end{aligned}$$

and, with a and b interchanged,

$$b\delta(a) + \delta(b)a = d(ba) + 2\eta ba = 0.$$

This proves the converse of Result 3.2.

For the second assertion, suppose

$$\delta(x) = d_1(x) + \eta x = d_2(x) + x\eta,$$

where d_1, d_2 are star derivations. Let $ab^* = b^*a = 0$. Use the first representation of δ in the first required identity. Since $d_1(b)^* = d_1(b^*)$, we obtain

$$\begin{aligned} a\delta(b)^* + \delta(a)b^* &= a(d_1(b) + \eta b)^* + (d_1(a) + \eta a)b^* \\ &= ad_1(b^*) + d_1(a)b^* + ab^*\eta^* + \eta ab^* \\ &= d_1(ab^*) = 0. \end{aligned}$$

Use instead the second representation in the other required identity:

$$\begin{aligned} \delta(b)^*a + b^*\delta(a) &= (d_2(b) + b\eta)^*a + b^*(d_2(a) + a\eta) \\ &= d_2(b^*)a + b^*d_2(a) + \eta^*b^*a + b^*a\eta \\ &= d_2(b^*a) = 0. \end{aligned}$$

Thus Result 3.3 also has its converse. The calculation used no property of η beyond belonging to the ambient bimodule algebra, proving the stated strengthening. $\square$

4 Boundary checks and scope

The proof is valid for nonunital A : it never inserts a unit or an approximate identity. It is also genuinely bidual-valued; every product is taken in A^{**} , where the canonical copy of A acts associatively. The centrality in (R3.2) is essential to the displayed short proof, whereas the real-part centrality in (R3.3) is not needed for sufficiency.

There is no circular use of Proposition 3.4. Only the definitions of a derivation and a star derivation, the structural formulas in the conclusions, and the stipulated zero products are used. Thus the special assumptions in Proposition 3.4—von Neumann or unital simple algebra and A -valued range—play no role in the converses asked for.

5 Novelty audit

Exact-title and exact-question searches, the local run indexes, and an OpenAlex audit of the 21 works indexed as citing the source through 17 August 2026 found no paper stating this resolution. The citing works concern related zero-product, Lie-centralizer, and anti-derivable-map problems. This supports, but cannot certify, novelty. Because the proof is elementary and the source wording may simply reflect an oversight, priority is not claimed.

References

- [1] B. Fadaee and H. Ghahramani, *Linear maps on C^* -algebras behaving like (anti-)derivations at orthogonal elements*, Bull. Malays. Math. Sci. Soc. 43 (2020), 2851–2859; arXiv:1907.03594.